THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assessment, 2025

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 25 pages (including this page) with 8 questions and 90 total marks

Permitted Materials

- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. There is extra space you can use for answers to any question commencing on page 22. If you still do not have enough space, tick the box near the bottom of page 25 and request extra pages from an invigilator—include the question number at the top of each additional page.
- You must NOT remove this question paper, or any extra pages provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request extra pages if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with extra pages if any, at the conclusion of the examination.

©University of Melbourne 2025

Can be placed in Baillieu Library

Blank page

Question 1 (9 marks)

Let p and q be primitive statements.

(a) Write a truth table for the compound statement

$$q \implies (p \vee (\sim q)).$$

(b) If q is true and $q \implies (p \vee (\sim q))$ is true, what can you conclude about the truth value of p? Justify your answer in terms of specific lines in truth table from Part (a).

(c) Find a statement with **exactly one connective** $\sim$, $\wedge$, $\vee$, $\implies$, or $\iff$ which is logically equivalent to $q \implies (p \vee (\sim q))$. Justify your answer using a truth table.

Question 2 (9 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists 1/x \in \mathbb{R} \setminus \{0\} \quad x \cdot 1/x = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Using only the Axioms of the Real Numbers (and rules of logic) together with the fact that $0x = 0$ for all $x \in \mathbb{R}$, write a (semi-)formal proof showing that for all $a, b \in \mathbb{R}$

$$\text{if } a > 0 \text{ and } b < 0, \text{ then } ab < 0.$$

Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 3 (9 marks)

Let A and B be subsets of $\mathbb{R}$ which are both non-empty and bounded above. Define

$$A + B = \{x + y : x \in A \text{ and } y \in B\}.$$

(a) Show that if α is an upper bound of A and β is an upper bound of B, then $\alpha + \beta$ is an upper bound of $A + B$.

(b) Show that $\sup(A + B) = \sup A + \sup B$.

Question 4 (12 marks)

For each of the following situations, provide a specific example for which the situation is true, and briefly justify your answer.

(a) A set $E \subseteq \mathbb{Q}$ such that the set of all limit points of E is $[0, \infty)$.

(b) A sequence (f_n) such that (f_n^2) converges to 9 but (f_n) diverges.
(Here $f_n^2 = f_n \cdot f_n$ for each $n \in \mathbb{N}^+$.)

(c) A convergent series $\sum_{n=1}^{\infty} a_n$ such that $\sum_{n=1}^{\infty} a_n^2$ diverges.

Question 5 (13 marks)

Let (x_n) be the sequence such that $x_1 = 0$ and

$$x_{n+1} = \frac{9}{6 - x_n}$$

for each $n \in \mathbb{N}^+$.

(a) Using a proof by induction, show that $0 \leq x_n < 3$ for all $n \in \mathbb{N}^+$.

(b) Is the sequence (x_n) monotone increasing, monotone decreasing, or neither? Justify your answer.

(Hint: Compute $x_{n+1} - x_n$.)

(c) Is the sequence (x_n) Cauchy? Briefly justify your answer.

Question 6 (11 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 2x - 4 & \text{if } x < 2 \\ \dfrac{x}{3x+1} & \text{if } x \geq 2 \end{cases}$$

(a) Using the $\epsilon - M$ definition of limits, show that $\lim_{x \to \infty} f(x) = 1/3$.

(b) Using the $\epsilon - \delta$ definition of continuity, show that f is not continuous at 2.

Question 7 (13 marks)

Let $f : [-1, 1] \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 1 - x^2 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

(a) Let $P = \{x_0, x_1, x_2, \ldots, x_n\}$ be any partition of $[-1, 1]$. Compute the lower Riemann sum $L(f, P)$.

(b) Find a positive lower bound for the upper Riemann integral of f on $[-1, 1]$.

(c) Is f Riemann integrable on $[-1, 1]$? Justify your answer using Parts (a) and (b).

Note. *For Question 8, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 8 (14 marks)

(a) For each of the following infinite series, use an appropriate test or theorem to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{2n}{n^4 + 16}$

(ii) $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n \log n}$

(b) Find the radius of convergence and interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{3^n x^n}{\sqrt{n(n+1)}}$$

Additional writing space for any question commences on the next page

Additional answer space for any question—submit this page even if blank

Additional answer space for any question—submit this page even if blank

Additional answer space for any question—submit this page even if blank

Additional answer space for any question—submit this page even if blank

End of Exam — Total Available Marks = 90

You must tick this box if you have used extra pages ☐